

1.1 What Is This About?

In the pages to follow, you'll see a lot of years and months and a lot of events. They seem isolated. What picture do they attempt to paint? What are we getting at? In fact, what is this book about?

The simple answer would be that it is a history of computing. "History is bunk," a notable American once said, and we couldn't agree less. To that's man's credit, he said it in a different context. Here, we're talking about what brought us to where we are, insofar as we are technology consumers and enthusiasts.

The complex answer is, well, slightly more complex: all of computing, information technology, computation, etc. is based upon one thing—the manipulation of ones and zeroes, or bits. It is all just that—whether it's the multiplying of two numbers, as was attempted on machines four hundred years ago, or the idea of meeting a schoolmate on Orkut. How?

Well, there are human needs, and computing happens to be one solution to those needs. To emphasise, *many* human needs can be dealt with by computation. So the computing—playing with those bits—is the base, and our needs and solutions are placed upon that foundation.

To tie these two together, what we have is a situation where *how* bits are manipulated dictate how some (in fact, many) human needs are handled. That is the path we have taken. So, for example, when you "meet" that friend on Orkut, what's happening? First you see his picture on the site. That picture is made of bits. The site itself is bits on a server. Then you send him a message—those are bits travelling over the fibre or telephone line. And so on. Now what's interesting here is the following:

- (a) Bit-manipulation is the beginning; we go on to transistor manipulation, then to integrated circuit manipulation, and so on. Structure upon structure, vertically; branching out hori-